



Water footprint and crop water productivity as decision-support indicators for water resources management and sustainable agricultural planning in Arauca, Colombia

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Abstract

Agricultural expansion in tropical frontier regions is increasing pressure on water resources and creating new challenges for sustainable water-use planning and management. In Colombia, however, local-scale assessments that integrate water-use indicators into agricultural decision-making remain limited, particularly in departments such as Arauca, where productive expansion coexists with growing pressure on water availability. This study assessed the green water footprint, defined as rainwater stored in the soil and consumed by crops, and the blue water footprint, defined as surface and groundwater consumed through irrigation, for eight regionally relevant crops in Arauca, Colombia, during 2023: rice, maize, cacao, papaya, passion fruit, pineapple, citrus, and plantain. Crop water productivity was integrated as a complementary indicator to support water-resource-based agricultural planning at the municipal level. The assessment combined the Water Footprint Network methodology with crop productivity estimates derived from crop yield and water use. The crop-level unit water footprint results showed a slightly greater contribution of blue water (52%) than green water (48%) across the evaluated crops, indicating considerable irrigation dependence in several production systems. Rice, cacao, and plantain exhibited a greater contribution from blue water, whereas pineapple and maize were more strongly supported by green water. The highest water productivity values were observed in Fortul, Saravena, and Arauquita, suggesting more favorable conditions for efficient water use in these municipalities. Overall, the findings reveal marked spatial differences in water-use patterns, irrigation dependence, and crop performance across the department. These results suggest that combining water footprint and crop water productivity can provide useful decision-support information for water resources management, crop prioritization, and territorially differentiated agricultural planning in tropical regions.

Keywords Water footprint · Crop water productivity · Sustainable agricultural planning · Sustainable water-use planning · Water resources management

Introduction

Agriculture remains one of the dominant freshwater-withdrawing sectors worldwide, although the commonly cited estimate that agriculture accounts for nearly 70% of global

freshwater withdrawals should be interpreted with caution because its empirical support and global representativeness have been questioned (Puy et al. 2025). Under these conditions, the challenge is no longer simply to increase food production, but to do so in ways that preserve hydrological sustainability, ecosystem integrity, and long-term territorial resilience (Wang 2025; Billi et al. 2025). This challenge is particularly acute in seasonally water-limited tropical agricultural frontier regions, where productive expansion often outpaces the planning instruments needed to integrate water availability, rainfall seasonality, ecosystem constraints, and crop suitability into territorial decision-making (FAO 2024; Cotrina-Sanchez et al. 2026). In these regions, rainfall

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seasonality and interannual climate variability, including ENSO-related anomalies, may alter the balance between green and blue water use by modifying effective rainfall, crop water requirements, and irrigation demand (Giraldo-Osorio et al. 2022). Among the indicators used to evaluate agricultural water use, the water footprint has become one of the most widely applied and methodologically robust approaches because it quantifies the freshwater consumed during crop production and distinguishes between green water supplied by effective precipitation and blue water derived from surface and groundwater used for irrigation (Hoekstra et al. 2011; Vargas-Pineda et al. 2020; Mialyk et al. 2024a; Demir and Muratoglu 2025; Apolo-Masache et al. 2026). Beyond its original role as an accounting framework, the water footprint has evolved into an analytical tool for characterizing spatial patterns of agricultural water demand, comparing crop systems across regions, and identifying opportunities for more sustainable water allocation and crop planning (Xu et al. 2025; Joseph et al. 2026). Recent reviews and global datasets further show that water footprint research is shifting from broad inventories toward spatially explicit and decision-oriented applications, thereby increasing its value for subnational planning and regional sustainability assessments (Demir and Muratoglu 2025; Chukalla et al. 2025).

However, volumetric water use alone is insufficient for agricultural planning because reductions or increases in total water use do not necessarily translate into proportional changes in yield, water productivity, or overall system performance (Hashemi et al. 2024; Wang et al. 2025). A crop may exhibit a relatively high water footprint and still deliver favorable output per unit of water consumed, whereas another may use less water in absolute terms but perform poorly in productive efficiency (Pacetti et al. 2025; Wang et al. 2025). For this reason, water footprint assessments gain stronger interpretive value when combined with crop water productivity, which links yield to consumptive water use and offers a more functional measure of how efficiently water is converted into agricultural output (Hashemi et al. 2024; Pacetti et al. 2025). Recent studies have shown that the joint use of water footprint and crop water productivity improves the assessment of agricultural sustainability and provides a stronger basis for decisions on crop prioritization, irrigation management, and the spatial organization of farming systems under differentiated water-use conditions and increasing water constraints (Ma et al. 2024; Hashemi et al. 2024; Wang et al. 2025; Chen et al. 2025).

This integrated perspective is especially relevant in the Colombian Orinoquia, which is increasingly recognized as one of the country's major agricultural frontiers, where unplanned expansion may intensify pressures on biodiversity, fragile ecosystems, and water resources. In the

eastern high plains of the Orinoco, previous regional studies have shown that agricultural growth without orderly and evidence-based territorial planning may deepen tensions between production objectives and conservation priorities (González-Orozco et al. 2023; Del Río Duque et al. 2025). Within this broader regional context, Arauca represents a strategic case. Official planning data indicate that 1,643,161 ha of the department fall within Colombia's national agricultural frontier, equivalent to 68.9% of its territory. In 2022, Arauca reported 97,675 ha planted with agricultural crops, with plantain, cacao, rice, and maize among the most important in terms of cultivated area. Official land-suitability assessments further identify substantial potential for crops such as rice, cacao, maize, and papaya, confirming that Arauca is not only agriculturally relevant, but also highly sensitive to how water-dependent cropping systems are promoted and spatially distributed (UPRA 2023a; 2023b).

Despite this strategic importance, the Colombian literature on agricultural water footprint remains geographically uneven. Previous studies in the Colombian Orinoquia have addressed water-footprint assessments for specific crops, including oil palm and rice; however, these crops are cited here only as part of the regional literature background and do not define the crop set evaluated in the present study. The national evidence base remains spatially fragmented, and Arauca is still only marginally represented (Tovar-Hernández et al. 2017; Trujillo-González et al. 2022; Góngora-Duarte and Trujillo-González 2025). This gap is particularly critical in regions where agricultural planning should respond closely to water constraints, irrigation dependence, and long-term territorial sustainability. At the same time, Colombian planning instruments, particularly the 2022 National Water Study and UPRA's water-based territorial planning framework, have increasingly emphasized the need to align productive development with water availability and irrigation planning (IDEAM 2023; UPRA 2021). In this context, generating crop-specific and municipality-level evidence on water footprint and water productivity is not merely a technical exercise, but a planning requirement for improving crop prioritization and the spatial allocation of farming systems under limited water availability (Si et al. 2025; Demir 2026).

Accordingly, this study assessed the green and blue water footprint of eight regionally relevant crops in the Department of Arauca, Colombia, for 2023 and integrated crop water productivity as a complementary indicator to support territorial agricultural planning. The novelty of the study lies not in proposing a new water-footprint indicator, but in combining established water-use metrics at the municipal scale to identify spatial differences in irrigation dependence, hydro-agronomic efficiency, and crop-specific planning

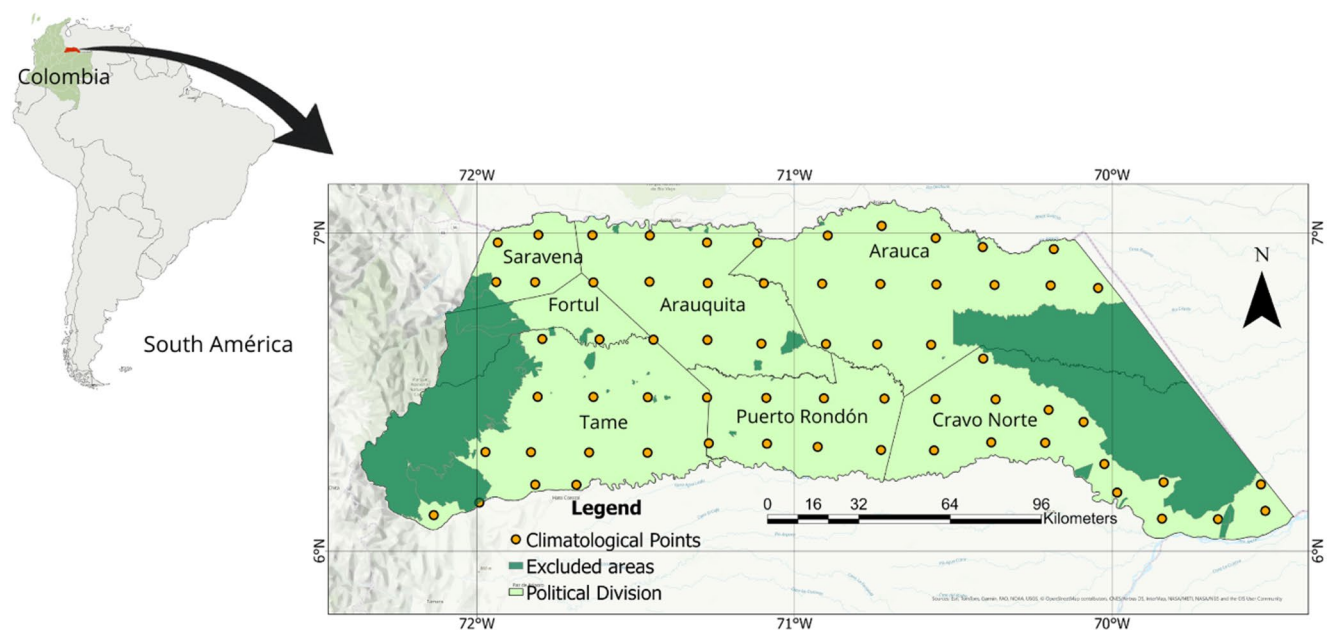


Fig. 1 Location of the study area in Arauca Department, Colombia, showing political-administrative divisions and climatological points used for climate data extraction

Table 1 Mean climatic conditions of the studied municipalities in Arauca, Colombia

Municipality	MAP (mm)	Tmax (°C)	Tmin (°C)
Tame	2013.9	29.2	15.0
Fortul	2280.5	34.5	18.5
Saravena	2189.9	35.1	18.5
Arauquita	2124.6	38.4	21.2
Arauca	2023.3	40.0	21.5
Cravo Norte	2149.6	40.0	21.7
Puerto Rondón	1760.8	39.0	21.5

MAP, mean annual precipitation; Tmax, mean maximum temperature; Tmin, mean minimum temperature

criteria in a tropical agricultural frontier that remains under-represented in national water-footprint assessments.

Materials and methods

Study area

This study was conducted in the Department of Arauca, located in the Colombian Orinoquia, and covered the municipalities of Arauca, Arauquita, Cravo Norte, Fortul, Puerto Rondón, Saravena, and Tame (Fig. 1). Arauca represents an important agricultural frontier in eastern Colombia and has experienced sustained expansion of regionally and commercially important crops. According to official agricultural statistics, the department reported 97,675 ha planted in 2022, with plantain, cacao, rice, and maize among the most representative crops in terms of planted area. The department

exhibits a monomodal rainfall regime, characterized by one main rainy season from April to November and one dry season from December to March, associated with the seasonal displacement of the Intertropical Convergence Zone.

Because this study was intended to support water-resource-based agricultural planning, areas subject to legal or environmental restrictions for agricultural use, including national natural parks, indigenous reserves, protective forest reserves, and private natural reserves, were excluded from the analysis. These restrictions were incorporated into the GIS workflow by overlaying official protected-area and land-restriction layers and applying a spatial mask to exclude all restricted areas from the analytical domain.

Climatic, soil, and crop data

Climatic variables required to estimate crop water requirements, including maximum and minimum temperature, relative humidity, wind speed, precipitation, and sunshine duration, were obtained from the NASA POWER database for 2023 (NASA POWER 2026) (Table 1). NASA POWER is a gridded database that provides solar and meteorological data derived from satellite observations and atmospheric models for applications in energy, infrastructure, and agroclimatology. Its meteorological products are mainly based on NASA's MERRA-2 reanalysis/assimilation system and are currently available at a native spatial resolution of $0.5^\circ \times 0.625^\circ$ approximately 30×37.5 arc-min. To represent climatic variability across the department, 65 centroid points were distributed over the study area and used for data

extraction (Fig. 1). These points were intended to sample the spatial climatic gradient at the departmental scale; however, the effective representativeness of the extracted values is constrained by the native resolution of the NASA POWER meteorological grid. Therefore, the climatic inputs were considered appropriate for a municipality-level regional assessment, but not for site-specific microclimatic interpretation. The climatic variables were used as fixed inputs for the 2023 baseline assessment to ensure consistency with crop-yield and municipal agricultural data for the same reference year, rather than as long-term stationary climate conditions.

Soil physicochemical and hydrological properties were derived from the IGAC soil survey for Arauca Department (IGAC 2012). The parameters considered included total available water, maximum infiltration rate, maximum rooting depth, and initial soil moisture depletion. These variables were processed using SPAW Hydrology to improve the characterization of soil water storage and movement under crop conditions (Saxton and Rawls 2006; Saxton et al. 2006).

Eight crops were selected based on their regional relevance in official agricultural statistics and their projected expansion within the agricultural frontier: rice, maize, cacao, papaya, passion fruit, pineapple, citrus, and plantain (UPRA 2023^a). Plantain, cacao, rice, and maize are among the most representative crops in Arauca in terms of planted area, while official land-suitability assessments identify additional potential for crops such as rice, cacao, maize, and papaya (UPRA 2023^a; b). Passion fruit, pineapple, and citrus were also included because they are reported in the Municipal Agricultural Assessments and could therefore be evaluated within the same municipality-level data framework. Crop yield data were obtained from the Municipal Agricultural Assessments, whereas crop coefficients (K_c), crop-cycle duration, and crop calendars were assigned following FAO guidelines for crop evapotranspiration and irrigation planning, with adjustments according to regional production conditions (Allen et al. 1998). Based on these inputs, actual crop evapotranspiration (ET_a) was estimated for each crop and municipality as the accumulated

evapotranspiration over the simulated crop cycle using monthly climatic inputs for 2023. The resulting ET_a values are presented in Table 2 and should not be interpreted as long-term climatic averages. Therefore, the estimated ET_a values, irrigation requirements, and green–blue water partitioning represent the 2023 baseline conditions and should not be interpreted as multi-year averages or as long-term irrigation prescriptions. Interannual rainfall variability, ENSO-related anomalies, and changes in crop calendars or cultivated varieties may modify the timing and magnitude of effective rainfall, crop water requirements, and irrigation demand (Allen et al. 1998).

Estimation of green and blue water footprint

The water footprint was estimated following the methodology proposed by the Water Footprint Network, which defines the water footprint as the volume of freshwater consumed directly and indirectly during the production of goods and services (Hoekstra et al. 2011). In this study, the total water footprint (WF) was calculated as the sum of the blue water footprint (BWF) and the green water footprint (GWF) (Eq. 1), where WF is expressed in $\text{m}^3 \text{ton}^{-1}$.

$$WF [\text{m}^3 \text{ton}^{-1}] = GWF + BWF \quad (1)$$

The green and blue components were estimated from crop water use and crop yield as shown in Eqs. 2 and 3, where CWU_{green} and CWU_{blue} represent the green and blue components of crop water use ($\text{m}^3 \text{ha}^{-1}$), respectively, and Y denotes crop yield (ton ha^{-1}).

$$GWF [\text{m}^3 \text{ton}^{-1}] = \frac{CWU_{\text{green}}}{Y} \quad (2)$$

$$BWF [\text{m}^3 \text{ton}^{-1}] = \frac{CWU_{\text{blue}}}{Y} \quad (3)$$

Following the Water Footprint Assessment Manual, GWF , BWF , and TWF were reported here as unit water footprints,

Table 2 Accumulated actual crop evapotranspiration (ET_a , mm growing cycle⁻¹) estimated over the simulated crop growing cycle for the evaluated crops across municipalities in Arauca, Colombia, using monthly climatic inputs for 2023

Municipality	Cacao	Maize	Plantain	Papaya	Pass. fruit	Rice	Citrus	Pineapple
Tame	1882.5	492.1	2058.5	2058.5	1594.6	678.7	1540.3	545.0
Fortul	1647.7	421.7	1793.2	1793.2	1414.4	585.0	1347.8	647.3
Saravena	1729.8	494.4	1883.4	1883.4	1451.2	654.0	1414.1	655.0
Araquita	1850.3	493.1	2023.3	2023.3	1554.6	674.2	1512.9	684.5
Arauca	2193.4	513.7	2392.7	2392.7	1910.3	694.3	1796.2	867.6
Cravo Norte	2358.7	120.0	2574.1	2574.1	2079.2	698.3	1936.6	956.0
Puerto Rondón	1880.4	492.0	2056.4	2056.4	1588.7	677.6	1540.3	704.0

ET_a : actual crop evapotranspiration accumulated over the simulated crop growing cycle. Values were estimated using monthly climatic inputs for 2023 and should not be interpreted as long-term climatic averages

expressed in $\text{m}^3 \text{ton}^{-1}$, whereas *CWUgreen* and *CWUblue* represent crop water use expressed in $\text{m}^3 \text{ha}^{-1}$.

Crop water use was estimated using CROPWAT 8.0 following the FAO approach for crop water requirement and irrigation requirement assessment (Smith 1992; Allen et al. 1998). CROPWAT 8.0 is designed to calculate crop water requirements and irrigation requirements based on climatic, crop, and soil data. The procedure included four main steps. First, monthly climatic data were entered into the climate module to estimate reference evapotranspiration (ET_o) using the FAO Penman–Monteith method. Second, crop files were defined for each of the eight evaluated crops by assigning crop coefficients (K_c), growth-stage duration, rooting depth, and crop calendars according to FAO guidelines and regional production conditions. Third, soil parameters, including total available water, maximum infiltration rate, maximum rooting depth, and initial soil moisture depletion, were incorporated to represent soil-water storage and water availability during the crop cycle. Fourth, the crop water requirement and irrigation requirement modules were used to estimate crop evapotranspiration, effective rainfall, net irrigation requirement, and the partitioning of crop water use into green and blue components.

Crop evapotranspiration (ET_c) was estimated using Eq. 4:

$$ET_c [\text{mm growing cycle}^{-1}] = ET_o \times K_c \quad (4)$$

where ET_c denotes crop evapotranspiration, ET_o represents reference evapotranspiration, and K_c is the crop coefficient. Green crop water use was attributed to the fraction of crop evapotranspiration supplied by effective precipitation during the crop cycle, whereas blue crop water use corresponded to the fraction supplied by irrigation water when effective precipitation was insufficient to meet crop water requirements. Thus, *GWF* and *BWF* in this study correspond to unit water footprints expressed in $\text{m}^3 \text{ton}^{-1}$. They should not be confused with the water footprint of production, which refers to total consumptive water use expressed in m^3 or $\text{m}^3 \text{year}^{-1}$. Water depths estimated in millimeters were converted into volumetric crop water use using a factor of 10, where 1 mm over 1 ha equals $10 \text{ m}^3 \text{ha}^{-1}$. The resulting green and blue crop water use values were then divided by crop yield to estimate the green and blue water footprint components in $\text{m}^3 \text{ton}^{-1}$.

It is important to note that the blue water footprint estimated in this study represents consumptive blue water use, that is, the fraction of crop evapotranspiration supplied by irrigation water. Therefore, it should not be interpreted as gross irrigation withdrawal, since conveyance, distribution, and application losses were not included. Irrigation efficiency was not explicitly incorporated because reliable

crop- and municipality-specific irrigation efficiency data were not available for the study area.

Crop water productivity

Crop water productivity (*CWP*) was calculated as the ratio between crop yield and total crop water use during the growing season, following Zwart and Bastiaanssen (2004) (Eq. 5). *CWP* is expressed in kilograms per cubic meter (kg m^{-3}), where *Y* denotes crop yield and *CWU* represents total crop water use throughout the crop cycle. This indicator was included to complement the water footprint analysis by incorporating a hydro-agronomic efficiency dimension, expressed as yield per unit of consumptive water use. However, *CWP* was not used as a direct measure of economic or nutritional value, since crops differ substantially in market price, caloric content, and productive function.

$$CWP [\text{kg m}^{-3}] = \frac{(Y \times 1000)}{CWU_{\text{total}}} \quad (5)$$

Because *CWP* is expressed in kg m^{-3} and *TWF* is expressed in $\text{m}^3 \text{ton}^{-1}$, both indicators are inversely related through the unit conversion $TWF = 1000/CWP$ when calculated from the same total *CWU* and crop yield.

The CROPWAT 8.0 simulations were based on three main groups of inputs: (i) climatic variables (maximum and minimum temperature, relative humidity, wind speed, sunshine duration, and precipitation) obtained from NASA POWER (NASA POWER 2026); (ii) crop parameters (crop coefficient, crop calendar, and crop-cycle duration) derived from FAO guidelines and adjusted to regional conditions (Allen et al. 1998); and (iii) soil properties (total available water, infiltration rate, rooting depth, and initial soil moisture) obtained from soil surveys and processed using SPAW Hydrology (IGAC 2012; Saxton and Rawls 2006; Saxton et al. 2006). Model outputs included reference evapotranspiration (ET_o), crop evapotranspiration (ET_c/ET_a), effective rainfall, and net irrigation requirements, which were used to estimate green and blue crop water use. These values were subsequently used to calculate green, blue, and total water footprints, as well as crop water productivity.

A detailed summary of the main input variables, units, data sources, processing steps, and derived outputs used to estimate water footprint and crop water productivity is provided in Supplementary Table S1. In addition, a crop- and municipality-level summary of the main calculated indicators is provided in Supplementary Table S2.

Spatial analysis

The spatialization and mapping of crop water productivity were performed using ArcGIS 10.1 (Esri, Redlands, CA, USA). The original spatial calculation unit corresponded to the 65 climatological centroid points used for climate data extraction. Crop water-use estimates were first generated at these points and then assigned to the municipality in which each point was located. For each crop, point-level values within the same municipality were averaged to obtain municipal crop water-use estimates, which were subsequently combined with municipal crop-yield data to calculate crop water productivity. The resulting municipal values were classified into discrete intervals to facilitate comparison and mapped while preserving municipal boundaries.

Analytical approach for sustainable water-use planning

The results were interpreted jointly using the green and blue water footprint components and crop water productivity at the municipal level (Mialyk et al. 2024b; Ma et al. 2024; Demir and Muratoglu 2025). Municipalities with a greater contribution of green water and higher crop water productivity were interpreted as having lower direct dependence on irrigation water under the 2023 baseline conditions, rather than as areas free of water-related risk (Falkenmark and Rockström 2006; Letseku and Grové 2022). This distinction is important because green water is also subject to scarcity, particularly when rainfall seasonality, soil-water storage, land-use change, and climate variability affect root-zone moisture availability (Rosa et al. 2020; Wang-Erlandsson et al. 2022). In contrast, municipalities with higher dependence on blue water and lower water productivity were interpreted as territories requiring stricter planning, irrigation-efficiency measures, or greater caution in the promotion of water-demanding crops (Pellicer-Martínez and Martínez-Paz 2016; Hashemi et al. 2024). This integrative approach was used to derive differentiated recommendations

for differentiated hydro-agronomic recommendations under contrasting water-use conditions and to support sustainable water-use planning in the Department of Arauca (Pellicer-Martínez and Martínez-Paz 2016; Ma et al. 2024).

Results and discussion

Overall pattern of agricultural water use in Arauca

The assessment of the unit water footprint of eight promising crops in the Department of Arauca showed a near-balanced distribution between green and blue water components. Across the crop-level unit water footprint values, green water accounted for approximately 48% and blue water for approximately 52% of the cumulative unit water footprint represented in Table 3. In water-footprint terminology, green water corresponds to rainwater stored in the soil and consumed by crops, whereas blue water refers to surface and groundwater consumed through irrigation (Rost et al. 2008; Mialyk et al. 2024a; Demir and Muratoglu 2025). These values indicate that the evaluated crop set was supported by both rainfall-derived soil moisture and irrigation-derived water, with a slightly greater contribution from blue water.

The crop-level results showed strong contrasts in both total water footprint and the relative contribution of green and blue water (Fig. 2). Cacao exhibited by far the highest total water footprint ($29,841.7 \text{ m}^3 \text{ ton}^{-1}$), with a very large blue-water component ($17,248.8 \text{ m}^3 \text{ ton}^{-1}$), whereas pineapple had the lowest total water footprint ($560.7 \text{ m}^3 \text{ ton}^{-1}$), almost entirely supported by green water. Passion fruit showed an intermediate total water footprint ($1,018.8 \text{ m}^3 \text{ ton}^{-1}$), with a greater contribution from green water ($648.2 \text{ m}^3 \text{ ton}^{-1}$) than from blue water ($370.6 \text{ m}^3 \text{ ton}^{-1}$). Rice also depended strongly on blue water, with $1048.0 \text{ m}^3 \text{ ton}^{-1}$ out of a total of $1201.4 \text{ m}^3 \text{ ton}^{-1}$, whereas maize was dominated by green water ($3118.6 \text{ m}^3 \text{ ton}^{-1}$ green vs. $60.2 \text{ m}^3 \text{ ton}^{-1}$ blue) but still showed a relatively high total footprint because of its large green-water component. Maize should therefore be interpreted as a crop with low irrigation dependence, rather than as a crop with a low total water footprint (Mialyk et al. 2024a; Demir and Muratoglu 2025).

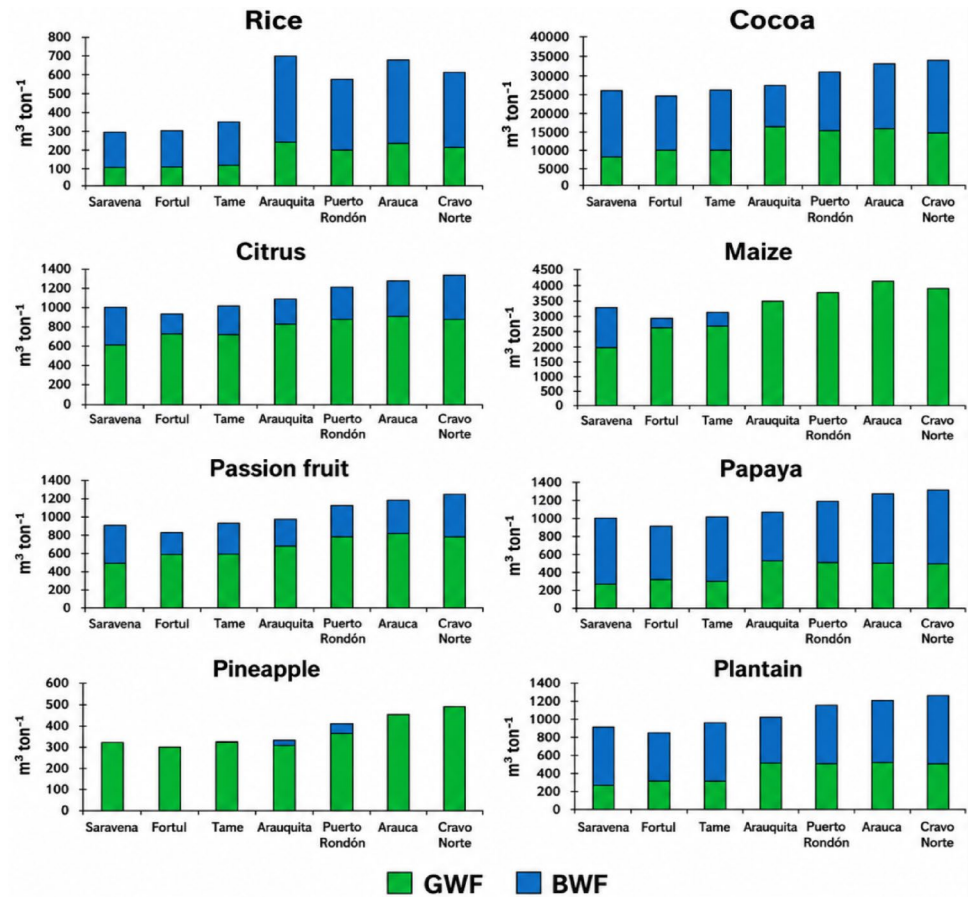
The relationship between annual precipitation and the green water footprint should be interpreted with caution. MAP values in Table 1 describe annual rainfall at the municipal scale, whereas *GWF* values in Table 3 correspond to unit green water footprints expressed in $\text{m}^3 \text{ ton}^{-1}$. Therefore, *GWF* is not a direct volumetric expression of annual rainfall, but the result of the fraction of crop water use supplied by effective precipitation during the simulated crop cycle divided by crop yield. Differences between MAP

Table 3 Green, blue, and total unit water footprints of the evaluated crops in Arauca Department, Colombia ($\text{m}^3 \text{ ton}^{-1}$)

Crop	GWF	BWF	TWF
Pineapple	551.4	9.2	560.7
Pass. fruit	648.2	370.6	1018.8
Plantain	416.0	674.1	1090.1
Citrus	789.8	360.6	1150.4
Papaya	451.5	727.1	1178.7
Rice	153.4	1048.0	1201.4
Maize	3118.6	60.2	3178.7
Cacao	12592.9	17248.8	29841.7

GWF, green water footprint; *BWF*, blue water footprint; *TWF*, total water footprint. Values correspond to unit water footprints and should not be interpreted as total annual water consumption

Fig. 2 Green and blue water footprint components, and total water footprint, of eight crops evaluated in the Department of Arauca, Colombia, expressed in $\text{m}^3 \text{ton}^{-1}$. Note: GWF green water footprint; BWF blue water footprint.



and *GWF* are consequently influenced by rainfall seasonality, crop calendars, evapotranspiration demand, soil-water storage, and yield levels (Hoekstra et al. 2011; Mialyk et al. 2024a). This explains why crops grown in municipalities with substantial annual rainfall may still show a high blue-water component when effective rainfall does not coincide with crop water demand or when irrigation is required during critical growth stages.

These patterns are consistent with benchmark literature. Global assessments identify cocoa among the crops with relatively large water footprints, whereas pineapple, papaya, citrus, and maize generally exhibit lower average values than high-footprint perennial commodities (Mekonnen and Hoekstra 2011). In the Mekonnen and Hoekstra crop water-footprint database, reported global averages are approximately $255 \text{ m}^3 \text{ton}^{-1}$ for pineapple, $460 \text{ m}^3 \text{ton}^{-1}$ for papaya, $560 \text{ m}^3 \text{ton}^{-1}$ for citrus, and $1222 \text{ m}^3 \text{ton}^{-1}$ for maize; the same report also identifies cocoa among commodities with relatively large water footprints. Against this background, Arauca's pineapple remains within the same order of magnitude as global benchmarks, papaya and citrus are moderately above global averages, maize is clearly above the global benchmark despite being mostly green-water-based, and cacao stands out as the crop imposing the

greatest water-use burden in the department (Mekonnen and Hoekstra 2011; Mialyk et al. 2024b; Demir and Muratoglu 2025).

Rice deserves particular attention. In Arauca, rice showed a total water footprint of approximately $1201 \text{ m}^3 \text{ton}^{-1}$, of which nearly 90% corresponded to blue water. This total value is close to the global average reported for paddy rice, around $1325 \text{ m}^3 \text{ton}^{-1}$, but its composition is substantially more irrigation-dependent than the global average, where the blue component represents roughly 44% (Chapagain and Hoekstra 2011). This means that the main issue in Arauca is not simply that rice has a relatively high total water footprint, but that production is sustained by a much more irrigation-intensive water profile, which is precisely the type of condition that matters for sustainable water-use planning (Li et al. 2023; Yu et al. 2024).

From a water resources management perspective, these results indicate that agricultural planning in Arauca should not rely only on land suitability or crop yield potential, but also on the source of water sustaining production and the degree of irrigation dependence associated with each crop. The relatively high contribution of blue water suggests that water allocation, irrigation efficiency, and crop prioritization should be considered explicitly in future agricultural

expansion. At the same time, crops with a greater green-water contribution should not automatically be interpreted as low-impact options, because total water footprint and crop water productivity must also be considered jointly.

Crop water productivity and municipal contrasts

Crop water productivity adds a hydro-agronomic efficiency dimension to the interpretation of agricultural water use because it indicates how much harvested mass is produced per unit of consumptive water use (Zwart and Bastiaanssen 2004; Letseku and Grové 2022). However, because the evaluated crops differ substantially in market value, caloric contribution, productive function, and crop cycle, *CWP* values should not be interpreted as a direct ranking of the overall value of different crops. Instead, they should be used primarily to identify spatial contrasts in water-use efficiency within each crop and across municipalities. In this study, the highest *CWP* values were generally concentrated in Fortul, Saravena, and Arauquita, whereas lower values were found in Cravo Norte and, for some crops, Tame. These contrasts indicate differences in hydro-agronomic performance and can support the territorial targeting of irrigation-efficiency and crop-management interventions (Ma et al. 2024; Wu et al. 2024), but they should be complemented with economic, nutritional, social, and ecological criteria before defining crop substitution or investment priorities (Fig. 3)

Figure 4 presents a heatmap of crop water productivity across the studied municipalities, allowing a comparative visualization of both spatial and crop-specific variations and highlighting where each crop showed relatively higher or lower hydro-agronomic performance. The color gradient highlights differences in productivity levels and facilitates the identification of crops and municipalities with relatively higher or lower water-use efficiency. The comparison with the literature reinforces the planning message. As an external benchmark for irrigated rice systems, reported rice water productivity values in northern Iran range from about 0.8 to 1.8 kg m⁻³, placing the best-performing Arauca municipalities near the lower end of that interval and the weakest ones clearly below it (Pourgholam-Amiji et al. 2021). For maize, the estimated average water productivity in the study area was approximately 0.28 kg m⁻³, indicating relatively low efficiency in the conversion of water into biomass. This value is considerably lower than those reported in other regions under more optimized production systems. For instance, studies conducted in the US High Plains have documented water productivity values close to 2.0 kg m⁻³, reflecting more efficient water use under controlled agronomic and climatic conditions (Trout and DeJonge 2017; Mbava et al. 2020). Classic reviews further reinforce this contrast, indicating that irrigated maize and rice typically

exhibit substantially higher water productivity than the lowest levels observed in this study (Zwart and Bastiaanssen 2004). This tendency is also consistent with evidence from pineapple systems, where improved management has been associated with gains in water productivity under tropical production conditions (Coelho et al. 2024). Altogether, these findings suggest that the planning challenge in Arauca is not only to reduce dependence on blue water, but also to improve agronomic performance so that water use is translated more effectively into yield (Zwart and Bastiaanssen 2004; Trout and DeJonge 2017; Mbava et al. 2020). Thus, the heatmap should be interpreted as a screening tool to identify relative differences in crop water productivity, rather than as a stand-alone basis for crop substitution or long-term planning decisions.

To synthesize the municipal contrasts in crop water productivity, a heatmap was used as a comparative visualization tool.

Implications for territorial agricultural planning

Taken together, the water-footprint and water-productivity results support a differentiated territorial strategy for agricultural development in Arauca, as summarized in Table 4. More importantly, they provide a practical basis for water resources management by identifying where crop expansion may be promoted under relatively favorable water-use conditions and where greater caution is required because of stronger irrigation dependence or lower productive efficiency (Pellicer-Martínez and Martínez-Paz 2016; Ma et al. 2024). Municipalities such as Saravena, Arauquita, and Fortul combine a greater contribution of green water with comparatively higher crop water productivity, suggesting lower direct dependence on irrigation water under the 2023 baseline conditions. However, this should not be interpreted as absence of water-related risk, since green-water availability may also be affected by rainfall seasonality, soil-water storage, and climate variability (Wu et al. 2024; Cheng et al. 2025). In contrast, Tame, Arauca, Puerto Rondón, and especially Cravo Norte exhibit stronger dependence on blue water and lower or intermediate productivity, indicating that future agricultural expansion in these areas should be accompanied by stricter irrigation-efficiency measures, improved agronomic management, and more careful evaluation of water availability (Yang et al. 2023; Smith et al. 2024; Cheng et al. 2025).

From this perspective, the contribution of the study does not lie in establishing an absolute ranking among crops, but in identifying hydro-agronomic contrasts in water source dependence and yield per unit of consumptive water use (Table 4). The joint interpretation of water footprint and crop water productivity provides a decision-support framework

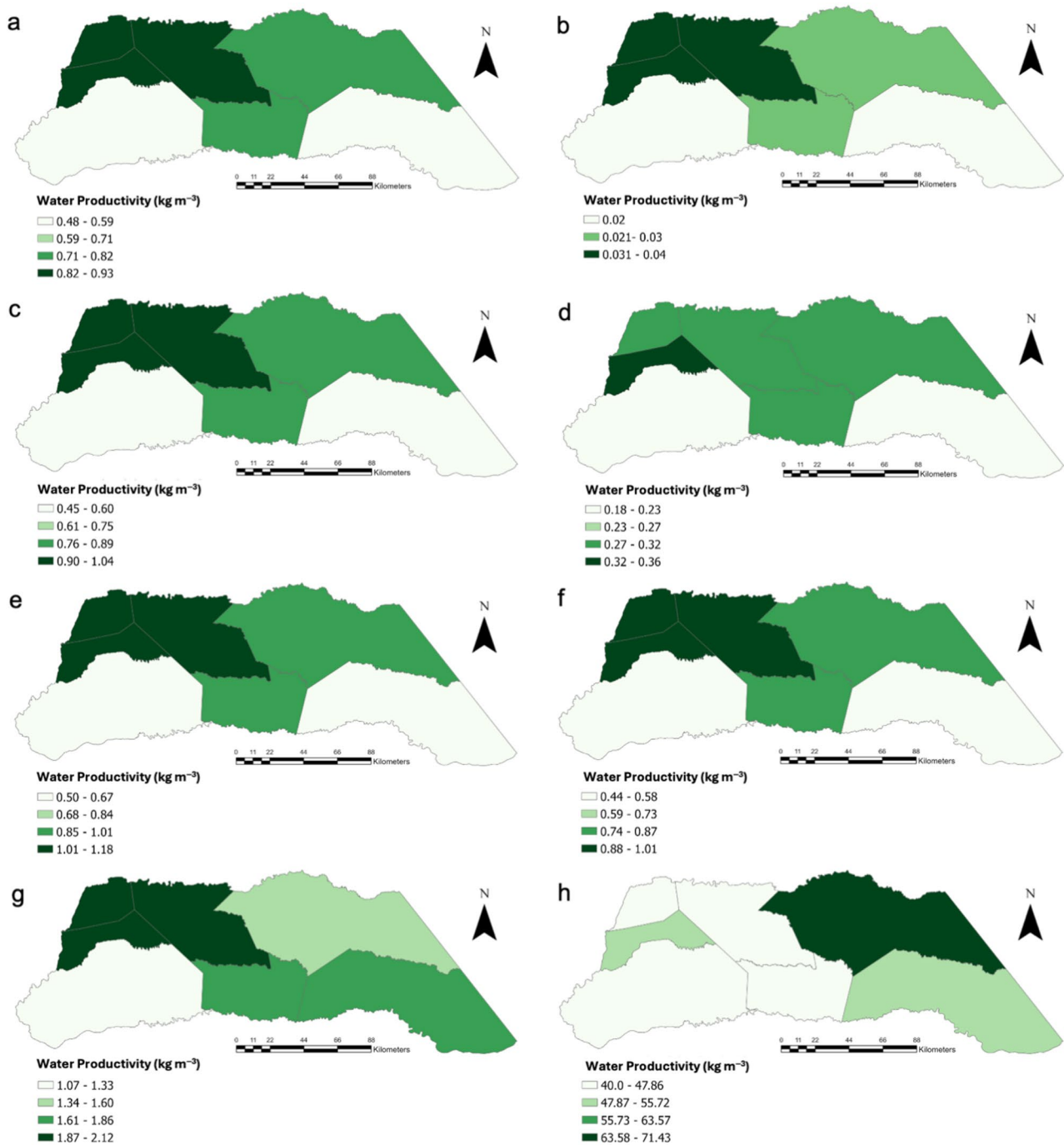


Fig. 3 Spatial distribution of crop water productivity for eight crops evaluated across the municipalities of Arauca, Arauquita, Cravo Norte, Fortul, Puerto Rondón, Saravena, and Tame in Arauca Department,

Colombia (kg m^{-3}): **a** rice, **b** cocoa, **c** citrus, **d** maize, **e** passion fruit, **f** papaya, **g** pineapple, and **h** plantain

for identifying territories where agricultural growth may be more compatible with sustainable water use and territories where expansion may increase pressure on water resources (Pellicer-Martínez and Martínez-Paz 2016; Ma et al. 2024). This is especially relevant in frontier regions such as Arauca, where agricultural development is advancing rapidly and planning instruments must anticipate potential

conflicts between productive expansion and long-term water sustainability (González-Orozco et al. 2023; Smith et al. 2024). Therefore, municipal-level differentiation should be considered as one input for agricultural planning and irrigation policy under water-constrained scenarios, alongside economic, nutritional, social, and ecological criteria (Wu et al. 2024; Cheng et al. 2025). From a practical planning



Fig. 4 Heatmap of crop water productivity for eight crops across seven municipalities in Arauca Department, Colombia, expressed in kg m^{-3} . Darker tones indicate higher relative crop water productivity within each crop

Table 4 Planning-oriented interpretation of water footprint and crop water productivity indicators in Arauca, Colombia

Municipality	Hydro-agronomic suitability	Water-use profile	Planning implication
Saravena	Pineapple, cocoa	High contribution of green water and high-water productivity	Promote rainfed-oriented systems and soil-moisture conservation practices.
Fortul	Pineapple, maize, papaya, citrus, cocoa	High water productivity and moderate water footprint	Maintain efficient production systems and restrict the expansion of water-intensive crops.
Tame	Maize, Pineapple	Moderate productivity and high dependence on blue water	Improve irrigation efficiency and crop yields.
Arauquita	Pineapple, maize, rice, papaya, cocoa	High green-water contribution and high-water productivity	Prioritize green-water-based crops and sustainable agronomic management.
Puerto Rondón	Maize, pineapple, papaya, passion fruit, citrus	Intermediate productivity and irrigation dependence	Improve agronomic practices and restrict the expansion of water-intensive crops.
Arauca	Plantain, maize, cocoa, papaya, pineapple	High dependence on blue water and relatively high productivity	Optimize irrigation scheduling and promote high-efficiency irrigation systems.
Cravo Norte	Pineapple, plantain, maize	Low productivity and high dependence on blue water	Combine irrigation efficiency with soil-moisture conservation and fertility management.

perspective, the proposed indicators should be understood as a screening tool rather than as direct policy prescriptions. Municipalities or crops with high blue water footprint values and low crop water productivity may require closer evaluation of water availability, irrigation scheduling, conveyance and application efficiency, and crop-management practices before promoting further expansion. Conversely, areas with lower irrigation dependence and relatively higher crop water productivity may represent more favorable conditions for water-efficient agricultural development,

provided that rainfall seasonality, soil-water storage, market demand, and ecological constraints are also considered. In this sense, the joint use of water footprint and crop water productivity can support municipal and regional planning by identifying where additional hydrological assessment, irrigation-efficiency programs, or more cautious crop-expansion decisions are needed (Hoekstra et al. 2011; Pellicer-Martínez and Martínez-Paz 2016; Ma et al. 2024; Smith et al. 2024). These planning implications should also be interpreted under the influence of climate variability. In

Arauca, changes in rainfall seasonality, dry-season intensity, or ENSO-related anomalies could modify effective rainfall and increase irrigation requirements (Giraldo-Osorio et al. 2022), particularly for crops whose water demand coincides with drier periods. Under such conditions, municipalities currently classified as less irrigation-dependent under the 2023 baseline may become more vulnerable in drier years (Giraldo-Osorio et al. 2022; Smith et al. 2024). Therefore, the proposed indicators should be updated with multi-year climatic data and complemented with climate-scenario analysis before being used for long-term agricultural zoning or irrigation investment decisions.

Study limitations

This study has some limitations that should be considered when interpreting the results. The single-year assessment and reliance on secondary and modeled data may not fully capture interannual climatic variability or field-scale differences in crop water use. Therefore, the results should be interpreted as a regional decision-support baseline rather than as a long-term agronomic prescription. Further validation using multi-year and field-based data would strengthen the assessment of the spatial patterns identified across the study area.

Conclusions

The results of this study have direct implications for water resources management and territorial agricultural planning in tropical frontier regions. By combining green and blue water footprint with crop water productivity, the study showed that agricultural development in Arauca should be guided not only by crop suitability or expected yield, but also by irrigation dependence and the efficiency with which water is converted into production. In this sense, municipalities with a greater contribution of green water and higher crop water productivity offer more favorable conditions for promoting water-efficient agricultural expansion, whereas municipalities with stronger dependence on blue water require stricter irrigation management, improved efficiency measures, and greater caution in the promotion of water-demanding crops.

The findings also demonstrate that water footprint and crop water productivity can serve as practical decision-support indicators for identifying spatial priorities, reducing potential pressure on water resources, and improving the territorial targeting of agricultural policies. The main scientific contribution lies in integrating these indicators at the municipal scale to reveal spatial contrasts in irrigation dependence and productive efficiency within an

underrepresented tropical agricultural frontier. This is especially important in regions such as Arauca, where productive expansion is occurring alongside increasing demands for sustainable water use. Integrating these indicators into agricultural planning and water management instruments could therefore strengthen the capacity of local and regional institutions to guide crop prioritization under differentiated water-use conditions and to support irrigation-related decisions under conditions of environmental and climatic uncertainty. Future research should extend the assessment to multiple years and incorporate field validation, water-scarcity indicators, and climate-variability scenarios.

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Code availability No custom code was developed for this study. All analyses were conducted using commercial and publicly available software, as described in the Materials and Methods section.

Declarations

Conflict of interest The authors declare no competing interests.

Ethics approval This study did not involve human participants, identifiable human data, or animals. Therefore, ethical approval was not required.

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